

PROFESSIONAL INSTALLATION MANUAL

Ubuntu Server 26.04 LTS
Virtual Machine Deployment & Initial Administration

Purpose: Provide a repeatable procedure that another system administrator can follow to install, configure, verify, and troubleshoot an Ubuntu Server virtual machine.

Prepared as a practical system administration guide

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1. Introduction

This manual documents a complete Ubuntu Server installation workflow based on the Ubuntu Server virtual-machine exercise completed in Oracle VM VirtualBox. The procedure covers installation media selection, network configuration, hostname and user creation, guided disk partitioning, OpenSSH installation, first login, and service verification.

Target result: a bootable Ubuntu Server system with hostname **server01**, a non-root administrative account, DHCP networking, a guided full-disk layout, and OpenSSH Server installed and running.

The demonstrated virtual machine used a 25 GB virtual disk and 2 GB RAM. The installation screen showed an Ethernet interface receiving the DHCP address **10.0.2.15/24**. Your environment may assign a different address.

Safety note: Guided – Use Entire Disk destroys existing data on the selected virtual disk. Confirm that the selected disk is the intended VM disk before accepting the storage configuration.

2. Ubuntu Server Installation Guide

2.1 Prepare the Virtual Machine

Create a new Ubuntu Server VM in VirtualBox. Allocate sufficient memory and CPU for the workload; the demonstrated VM used 2048 MB RAM, 1 CPU, and a 25 GB dynamically allocated virtual disk.

2.2 Attach the Ubuntu Server ISO

Open VM Settings → Storage. Select the optical drive and attach the Ubuntu Server ISO. Confirm that the ISO appears under the IDE optical device before starting the VM.

2.3 Start the Installer

Power on the VM and proceed through the Ubuntu Server installer using the keyboard. Select English as the language unless the assignment requires another language.

2.4 Keyboard Layout

Choose the appropriate keyboard layout, such as English (US), and continue.

2.5 Network Configuration

Allow the installer to use DHCP unless a static configuration is specifically required. Record the address displayed by the installer. In the demonstrated installation it was 10.0.2.15/24.

2.6 Proxy / Mirror

Accept the default package mirror settings unless your organization or lab provides a required proxy or mirror.

2.7 Installation Type

Choose the standard Ubuntu Server installation rather than the minimized option unless the lab explicitly requires minimized installation. Do not select third-party drivers unless they are required.

2.8 Storage

Choose **Guided – Use an entire disk**. Select the 25 GB VirtualBox disk. The demonstrated layout used an ext4 root filesystem with a separate /boot partition and LVM. Leave disk encryption disabled unless the exercise requires it.

2.9 Profile Setup

Create the administrator's non-root account. Enter the requested full name, username, and a strong password. Ubuntu grants administrative access through sudo rather than requiring direct root login.

2.10 Hostname

Set the server hostname to **server01**. Verify the spelling before continuing.

2.11 SSH Server

Enable **Install OpenSSH Server**. Do not install additional packages unless instructed.

2.12 Finish Installation

Allow the installer to copy files and configure the system. When prompted, reboot. Detach the ISO from the virtual optical drive if necessary so the VM boots from its virtual disk.

3. Initial Server Configuration

After reboot, log in using the non-root administrative username and password created during installation. Password characters are intentionally not displayed at the Linux console while typing; this is normal behavior.

3.1 Update package information and installed packages

```
sudo apt update
sudo apt upgrade
```

Review the proposed changes before confirming an upgrade. Reboot afterward if the system reports that a reboot is required.

3.2 Confirm the hostname

```
hostnamectl
```

The output should identify the configured hostname as **server01**.

3.3 Review network configuration

```
ip addr
ip route
```

Use these commands to identify the active interface, IPv4 address, and default route. DHCP addresses can change, so record the address when the assignment requires documentation.

3.4 Check SSH

```
sudo systemctl status ssh
```

The expected state after starting SSH is **active (running)**. In the demonstrated VM, SSH initially showed inactive and was then started with **sudo systemctl start ssh**.

3.5 Enable SSH at boot

```
sudo systemctl enable ssh
```

Enabling the service ensures SSH is configured to start automatically during normal boot. If SSH is already enabled, systemd will report that no change is necessary.

4. Server Verification Procedures

Check	Command	Purpose / Expected Result
Operating system	<code>cat /etc/os-release</code>	Confirms the installed Ubuntu release.
Kernel	<code>uname -r</code>	Displays the running Linux kernel version.
Hostname	<code>hostnamectl</code>	Confirms server identity and hostname.
IP address	<code>ip addr</code>	Shows network interfaces and assigned addresses.
Routing	<code>ip route</code>	Confirms a default route when network access is expected.
Disk usage	<code>df -h</code>	Checks mounted filesystems and available capacity.
Block devices	<code>lsblk</code>	Shows disks, partitions, and logical volumes.
SSH service	<code>systemctl status ssh</code>	Expected: active (running) after SSH is started.
Listening ports	<code>sudo ss -tulpn</code>	Shows services listening on TCP/UDP ports.

SSH verification example

```
sudo systemctl start ssh
systemctl status ssh
```

A successful result contains an **Active: active (running)** line and indicates that sshd is listening on port 22. If the service is inactive, use the troubleshooting section below.

5. Troubleshooting Guide

VM fails to boot / no operating system found

Open VirtualBox Settings → Storage and verify that the Ubuntu ISO is attached to the optical drive during installation. After installation, detach the ISO and ensure the virtual hard disk remains attached.

Installer cannot find the ISO

Confirm the ISO file is an actual Ubuntu Server ISO, not a folder containing extracted ISO contents. Reattach the ISO through the optical-drive selector.

No network address

Check VirtualBox Network settings. NAT is suitable for basic Internet access in a lab. In Ubuntu, run `ip addr` and `ip route` and verify the interface is up.

Cannot see password while typing

Linux does not echo password characters at the console. Type the password normally and press Enter.

SSH shows inactive (dead)

Start it with `sudo systemctl start ssh`, then run `systemctl status ssh`. If it should start automatically, run `sudo systemctl enable ssh`.

SSH fails to start

Inspect the detailed service log with `sudo journalctl -u ssh --no-pager`. Check configuration syntax and confirm that another process is not already using the required port.

Disk layout concern

Stop before accepting the storage summary if the wrong disk is selected. The demonstrated VM had a 25 GB virtual disk. Confirm the target disk size and device before formatting.

Permission denied for administration

Use the configured administrative account and prefix privileged commands with `sudo`. Do not enable direct root login unless specifically required by an approved policy.

6. Best Practices

- Use a descriptive hostname and document it consistently.
- Use a normal administrative account with sudo instead of routine root login.
- Use strong, unique passwords and never place passwords in documentation or screenshots.
- Keep the server updated with apt update and apt upgrade according to the lab or organization's maintenance schedule.
- Install only required packages to reduce the attack surface.
- Enable SSH only when remote administration is required. Prefer key-based authentication for production environments.
- Record IP addresses, disk layout, hostname, service state, and major configuration changes in an administration log.
- Before destructive storage operations, verify the target disk and maintain a backup or VM snapshot where appropriate.
- Use systemctl status and journalctl to verify services and diagnose failures rather than guessing.
- Separate lab configuration from production configuration. NAT and DHCP are convenient for a lab, while enterprise deployments may require controlled VLANs, static addressing, DNS, firewalls, monitoring, and centralized identity.

7. References

The following references provide authoritative installation and administration information. Access dates should be recorded according to your institution's documentation requirements.

- Ubuntu Server Documentation — Canonical. <https://documentation.ubuntu.com/server/>
- Ubuntu Server installation tutorial — Canonical. <https://documentation.ubuntu.com/server/tutorial/installation-tutorial/>
- Ubuntu OpenSSH documentation — Canonical.
<https://documentation.ubuntu.com/server/how-to/security/openssh-server/>
- systemd documentation — Freedesktop Project. <https://systemd.io/>
- Oracle VM VirtualBox documentation — Oracle. <https://www.virtualbox.org/manual/>
- Linux manual pages: systemctl, journalctl, ip, lsblk, df.

Administrator handoff checklist

- ISO attached and installation completed
- Hostname verified as server01
- Non-root administrator can log in
- Network address recorded
- Disk layout verified
- OpenSSH installed
- SSH service active and enabled
- Verification commands completed
- Screenshots/evidence captured where required
- Final configuration documented